

# Sagar Sinha

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## PROFESSIONAL SUMMARY

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Senior Backend Engineer/SDE III with 7+ years of experience designing and scaling distributed systems, microservices, and event-driven architectures in the FinTech domain. Expert in Java, Spring Boot, REST APIs, and cloud-native development on GCP, serving 3.5M+ users, processing \$1M+ transactions, and reducing deployment time by 50%. Skilled in system design, API standardization, and building secure, PSD2-compliant financial platforms.

## EXPERIENCE

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### Lloyds Technology Centre | Senior Software Engineer

Nov 2023 – Present

#### *Open Banking*

- Designed and developed open banking microservices serving **3.5M+ users** across Europe, ensuring 99.9% uptime under high transaction load.
- Built secure consent and Account Information Service (AIS) APIs fully compliant with **PSD2 and FAPI** regulations, enabling seamless third-party provider integrations.
- Architected unified cross-brand account servicing platform enabling **3M+ users** single-view access across multiple banking entities.
- Standardized REST API contracts across legacy banking systems, improving interoperability and reducing integration time by 30%.
- Deployed containerized microservices using **Docker and Kubernetes** on GCP, improving release stability and horizontal scalability.
- Led cross-functional design discussions across backend, security, and compliance teams; mentored 3 engineers on system design best practices.

### JP Morgan Chase & Co. | Software Engineer II

Jun 2019 – Sep 2023

#### *JPM Coin – Blockchain Payments Platform*

- Engineered Java/Spring Boot microservices powering cross-border blockchain settlements exceeding **\$1M per transaction** for institutional clients.
- Built multi-currency client onboarding APIs that enabled global expansion into 5+ new markets.
- Ensured ACID-level transaction reliability and data consistency across distributed financial systems using event-driven patterns.
- Mentored 5+ junior engineers, conducted technical interviews, and led code review processes to enforce quality standards.

#### *Global Distribution Engine – Mutual Funds Platform*

- Migrated** a monolithic dividend distribution system to a microservices architecture, reducing deployment time by **50%** and increasing release frequency 3x.
- Improved system performance and maintainability through modular service design, API versioning strategies, and database optimization.
- Designed rule-based REST APIs for calculating mutual fund dividend distributions, successfully onboarding **4 new institutional clients**.

#### *SCOUT – SLA Monitoring & Chargeback System*

- Developed validation and ETL processing modules for large-scale SLA datasets, ensuring accurate performance measurement across 50+ business units.
- Implemented chargeback calculation logic aligned with contractual SLAs, improving cost transparency and internal performance tracking.
- Optimized backend workflows and SQL queries, reducing manual reconciliation efforts by 40%.

## AGENTIC AI PROJECT

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### HirelyAI – AI-Powered Mock Interview Platform

[hirelyai.in](https://hirelyai.in)

Founder & Full-Stack Engineer

2025

- Built and launched a real-time conversational AI platform that helps candidates prepare for technical interviews, achieving **sub-800ms end-to-end voice latency**.
- Engineered a low-latency voice pipeline that captures microphone audio, transcribes it via Deepgram, generates AI interviewer responses using **Gemini 2.5 Flash**, and streams natural-sounding speech back via Cartesia.
- Implemented advanced optimizations including **speculative generation, sentence-level audio streaming, and filler phrase injection** to deliver fluid, human-like conversations.
- Designed full-stack architecture with TypeScript/React frontend, Express WebSocket server, and PostgreSQL persistence using Drizzle ORM.
- Deployed on **Google Cloud Platform (Cloud Run)** with containerized auto-scaling, supporting concurrent user sessions reliably.

**Tech Stack:** TypeScript, React, Node.js, Express, PostgreSQL, Drizzle ORM, Gemini API, Deepgram, Cartesia, Google Cloud TTS, WebSockets, Docker, GCP Cloud Run.

## EDUCATION

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### JSS Science & Technology University

Mysore, KA

Master of Computer Applications (MCA) – CGPA: 8.39/10

Aug 2017 – May 2019

## TECHNICAL SKILLS

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**Languages & Frameworks:** Java, Python, JavaScript, TypeScript, Spring Boot, Spring Cloud, Hibernate, React.js, Node.js, Express

**Architecture & Design:** Microservices, Monolith, REST APIs, Distributed Systems, Event-Driven Architecture, System Design, API Standardization, Scalability, High Availability, Design Patterns

**Cloud & DevOps:** GCP (Cloud Run, GKE), AWS (basics), Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions

**Databases & Messaging:** PostgreSQL, Oracle, MySQL, Redis, Apache Kafka, Data Modeling, SQL Optimization

**Authentication & Security:** OAuth2, JWT, OIDC, API Gateways, RBAC, PSD2, FAPI

**GenAI & Agentic Systems:** LLMs, OpenAI API, Gemini API, Claude Code, Vector Databases, Prompt Engineering, RAG, Embeddings, Model Context Protocol (MCP)

**Testing & Observability:** JUnit, Mockito, SonarQube, Test Coverage, Splunk, Dynatrace, Kibana, ELK Stack

**Developer Tools:** Git, GitHub, IntelliJ IDEA, VS Code, Postman, Bruno, Maven, Gradle

**Practices:** Agile, Scrum, TDD, Code Reviews, Release Management, Mentoring, Technical Interviewing